

Where Do LLMs Go Wrong? Diagnosing Automated Peer Review via Aspect-Guided Multi-Level Perturbation

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Abstract

Large Language Models (LLMs) are increasingly integrated into academic peer review, prompting debates between full automation and purely human evaluation. Emerging evidence suggests optimal peer review leverages both human expertise and AI capabilities, and several major conferences have already adopted AI-assisted reviewing practices. However, effectively integrating these reviewers requires an aspect-based understanding of LLM vulnerabilities, clearly identifying specific dimensions where AI is most prone to error. Prior studies broadly caution against LLM biases but lack precise, aspect-specific insights necessary for informed human-AI partnerships in peer-review processes. We propose an *aspect-guided, multi-level perturbation framework* to systematically diagnose LLM weaknesses in automated peer review. By introducing targeted perturbations across key review components (papers, reviews, rebuttals) and evaluating impacts along critical quality dimensions (contribution, soundness, presentation, tone, completeness), our framework functions as a diagnostic tool: deviations from expected rating shifts after perturbation directly reveal specific LLM vulnerabilities. Our empirical analyses uncover recurring weaknesses, including misclassification of methodological flaws, disproportionate influence of strong rejection recommendations, inadequate responses to incomplete or negatively toned rebuttals, and misinterpretation of incorrect critiques as rigorous evaluations. These vulnerabilities consistently persist across diverse prompting strategies and a broad set of widely-used LLMs (e.g., GPT-4o, Gemini 2.0, LLaMA 3). This diagnostic framework provides granular insights into LLM limitations, empowering conference organizers to establish pragmatic, aspect-specific guidelines and enabling balanced, informed, and robust peer-review practices.

CCS Concepts

• **Computing methodologies** → **Natural language processing.**

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1 Introduction

Academic peer review is essential to scientific validation, yet traditional reviewing processes are under growing strain from rapidly increasing manuscript submissions [3]. This surge intensifies challenges such as securing qualified reviewers, managing heavy workloads, and mitigating subjective biases [2]. To address these issues, major conferences, including AAAI and ICLR, have integrated Large Language Models (LLMs) into their peer review processes, leveraging AI to supplement initial reviews, summarize reviewer discussions, and enhance feedback quality [14, 24]. Such implementations highlight the potential of LLMs to notably enhance reviewing efficiency and consistency, with early studies showing substantial alignment between AI-generated and human-generated reviews [14, 24]. Encouraged by these initial successes, some advocate fully automating peer review, sparking vigorous debate about the balance between human and AI evaluation [17].

Despite promising early results, substantial concerns remain about the reliability, fairness, and transparency of LLM-based reviewing. Previous research has broadly cautioned against vulnerabilities such as hallucinations, biases, and adversarial sensitivity in LLM-generated content [6, 27, 28, 30]. However, these studies typically address such vulnerabilities in isolation, lacking precise, aspect-specific insights into where LLMs are most prone to errors—insights critical for effective human-AI integration into peer-review workflows.

To fill this critical gap, we propose an *aspect-guided, multi-level perturbation framework* (Figure 1) to rigorously diagnose and pinpoint LLM vulnerabilities within automated peer review. Specifically, our framework introduces targeted perturbations to three key peer-review components—papers, reviews, and rebuttals—each

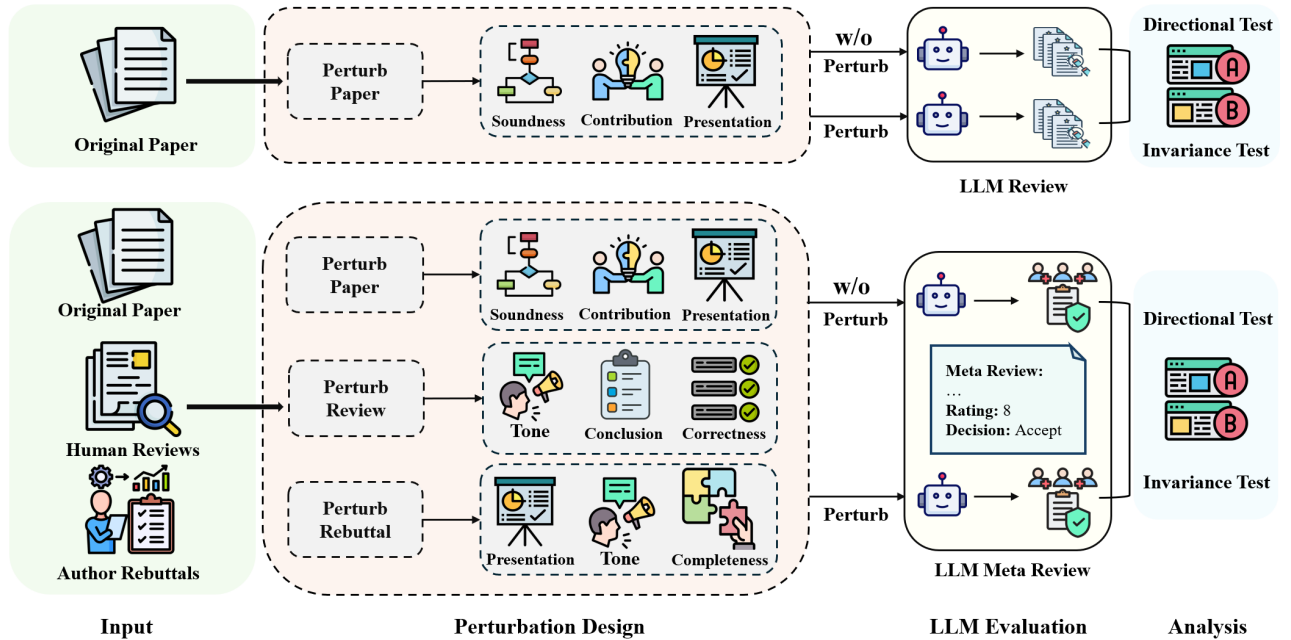


Figure 1: Overview of our aspect-guided, multi-level perturbation framework. We perturb one specific aspect of a single review component at a time—paper, reviews, or rebuttal—and evaluate the subsequent impact on LLM behaviors as reviewers or meta-reviewers.

along relevant quality dimensions: *contribution*, *soundness*, and *presentation* for papers; *tone*, *factual correctness*, and *conclusion* for reviews; and *tone*, *completeness*, and *presentation* for rebuttals. By perturbing a single aspect of one component at a time and carefully analyzing the subsequent effects on the behaviors of LLMs acting as reviewers or meta-reviewers, our approach reveals subtle yet consequential weaknesses, enabling precise identification of dimensions most susceptible to LLM biases.

We collect a representative dataset of 508 accepted papers from ICLR 2024 to empirically identify several recurring vulnerabilities inherent to current LLM-driven peer review. Notably, LLM reviewers frequently misclassify methodological flaws, confusing them with presentation issues; meta-reviewers are disproportionately swayed by strong negative recommendations; and incomplete or negatively toned rebuttals inadequately influence final decisions. Further, incorrect or misleading reviewer critiques are often misinterpreted as indicators of rigorous evaluation, unintentionally inflating assessments. Crucially, these vulnerabilities consistently persist across diverse prompting strategies—including Chain-of-Thought—and multiple widely-used LLMs such as GPT-4o, Gemini 2.0, LLaMA 3, Deepseek, and Mistral.

Our contributions are as follows:

- We develop a structured perturbation taxonomy and compile a comprehensive dataset containing 79,302 adversarial, aspect-guided perturbations spanning critical review components and dimensions.
- By applying our diagnostic framework, we empirically reveal several recurring vulnerabilities in current LLM-driven peer review,

such as misclassification of methodological flaws, disproportionate influence of strong negative recommendations, inadequate responses to incomplete or negatively toned rebuttals, and misinterpretation of incorrect critiques as indicators of rigor. These insights explicitly pinpoint review dimensions most susceptible to LLM-induced biases and errors.

- We provide a structured, reusable, and automated diagnostic toolkit that enables fine-grained, aspect-specific identification of vulnerabilities in LLM-based peer-review processes.¹ These granular insights equip peer-review workflow designers and conference organizers with critical evidence to formulate informed, aspect-specific policies for balanced and robust human-AI collaboration in peer review.

2 Perturbation Framework

We present our framework for assessing the robustness of LLMs in automated peer review. By introducing targeted perturbations across key components—specifically the paper, reviews, and rebuttal—and along multiple quality dimensions, we evaluate the effects of these manipulations on the behaviors of LLMs serving as both reviewers and meta-reviewers. All perturbations are intentionally negative, designed to degrade input quality.

We define two distinct roles for the LLMs in our framework, mirroring the human roles in the peer review process:

¹Our diagnostic toolkit and full perturbation dataset will be released to the community to facilitate future research.

LLM-as-Reviewer. This LLM role, denoted as M_{rev} , processes only the paper text, p , and outputs dimension-specific scores (e.g., Contribution, Soundness, Presentation) plus a final recommendation: $M_{\text{rev}}(p) \rightarrow \text{LLM Review}$.²

LLM-as-Meta-Reviewer. This LLM role, denoted as M_{meta} , takes as input the paper, p , multiple reviews, $\{r_i\}$, and the rebuttal, b : $M_{\text{meta}}(p, r_1, \dots, r_k, b) \rightarrow \text{LLM Meta-Review}$. It produces an overall score and final decision, reflecting the integrated viewpoints of multiple reviewers and the authors' responses.

We next formalize our single-target perturbation approach and detail the specific perturbation strategies employed.

2.1 Single-Target Perturbation

To analyze how each component (paper, review, rebuttal) affects LLM-based evaluations, we introduce a *single-target* perturbation approach. This means that we modify only one of these components at a time. Let $T = (p, r_1, \dots, r_k, b)$ denote the original, unperturbed inputs (the paper, a set of reviews, and the rebuttal). We define three perturbation functions:

- 1) $F_{\text{paper}}(T)$: Modifies the paper, resulting in $(p', r_1, \dots, r_k, b)$.
- 2) $F_{\text{review}}(T)$: Modifies the reviews, resulting in $(p, r'_1, \dots, r'_k, b)$.
- 3) $F_{\text{rebuttal}}(T)$: Modifies the rebuttal, resulting in $(p, r_1, \dots, r_k, b')$.

Applying any of these functions, $F_\alpha(T)$, produces a perturbed input, T' . We then compare the LLM outputs for the perturbed input, o_{pert} , with the unperturbed baseline outputs, o_{base} . For paper perturbations, we evaluate both $M_{\text{rev}}(p')$ and $M_{\text{meta}}(p', r_1, \dots, r_k, b)$. For review or rebuttal perturbations, we keep the paper, p , constant and evaluate only the meta-reviewer: $M_{\text{meta}}(p, r'_1, \dots, r'_k, b)$ or $M_{\text{meta}}(p, r_1, \dots, r_k, b')$.

2.2 Perturbation Taxonomy

We introduce a perturbation taxonomy (Figure 2) designed to rigorously assess the robustness of LLMs against targeted manipulations of the core peer review components: paper, review, and rebuttal.

Paper-Level 1) The perturbation of contribution exaggerates or distorts the novelty and discoveries, 2) the perturbation of soundness removes or conceals essential technical details, 3) the perturbation of presentation introduces typographical errors or disorganized text that undermine the clarity of the paper.

Review-Level 1) The perturbation of tone injects hostility or negativity into reviewer comments, 2) the perturbation of factual correctness introduces false statements as criticisms, 3) the perturbation of conclusion changes a positive recommendation into a strong rejection.

Rebuttal-Level 1) The perturbation of tone makes the author's response hostile, 2) the perturbation of presentation degrades the clarity of the rebuttal through poor formatting or typos, 3) the perturbation of completeness omits substantive replies to major reviewer concerns.

2.3 Perturbation Impelementation

Our perturbation framework implementation manipulates essential dimensions across the review cycle. Specifically, perturbations are performed as following with human manual validation:

Paper perturbations involve prompting GPT-4o to exaggerate originality statements (contribution), omit crucial methodological details (soundness), and deliberately degrade readability through typos and awkward phrasing (presentation).

Review perturbations include prompting GPT-4o to shift tone negatively (tone), insert explicitly false claims sampled from a maintained per-paper false-claim bucket (factual correctness), and perform direct text replacements to invert acceptance verdicts (conclusion).

Rebuttal perturbations similarly rely on GPT-4o prompts to introduce dismissive and hostile tones (tone), degrade readability with textual errors (presentation), and replace detailed responses with superficial placeholders (completeness).

3 Perturbation Details

3.1 Paper Perturbation Aspects

Paper Contribution Perturbation. We overclaim or distort the paper's contribution to cast doubt on its genuine value. Concretely, we prompt GPT-4o to insert exaggerated originality statements (e.g., "We are the *first* to propose any method of this kind...") while removing citations to closely related work. This can cause a diligent reviewer to question the authenticity of these claims.

Paper Soundness Perturbation. We prompt GPT-4o to remove or disguise crucial methodological and technical details (e.g., omit validation procedures and statistical testing methods for the experiment), thus making the paper less logically thorough consistent. By systematically rephrasing soundness-related content in this way, we create confusion around the rigor and effectiveness of the approach and test whether an LLM reviewer penalizes such shortcomings.

Paper Presentation Perturbation. We diminish the paper's clarity and polish by prompting GPT-4o to insert typos, awkward phrasings, or run-on sentences throughout. For instance, "Thes reuslts demontrate minimal improovment" might replace a cleaner sentence. This tests whether the LLM-based reviewer duly penalizes poor grammar and readability.

3.2 Review Perturbation Aspects

Review Tone Perturbation. We prompt GPT-4o to transform a neutral or constructive review into one laden with negativity or dismissiveness. For instance, "The evaluation seems limited" becomes "The evaluation is fatally flawed." This tests if a more hostile tone sways an LLM-based meta-review.

Review Factual Correctness Perturbation. We maintain a per-paper "false claim bucket" of misleading or blatantly incorrect statements (e.g., "The authors used only 2 training samples," "No experiments were conducted"). In practice, we prompt GPT-4o to generate several false claims for each paper, store them, and then randomly sample three of these claims per reviewer. Then we insert these three falsehoods into the reviewer's "Weakness" section. This tests whether the meta-review automatically detects these inaccuracies.

²In practice, a human reviewer may also consult the rebuttal or engage in follow-up discussions, but some do not consistently do so. For simplicity and experimental tractability, our *LLM-as-Reviewer* setting only ingests the paper.

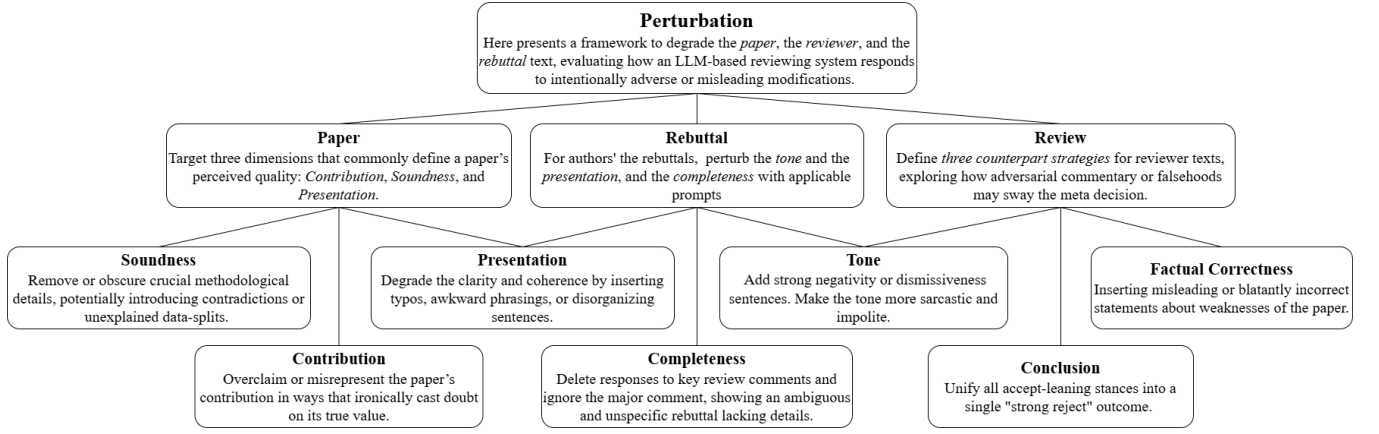


Figure 2: Taxonomy of perturbation strategies for altering the paper, review, or rebuttal text.

Review Conclusion Perturbation. Because our dataset comprises *accepted* papers (with generally positive reviews), we use a simple string-based replacement to flip any accept-leaning rating to a *strong reject*. Concretely, if a reviewer’s score or stance is “weak accept,” “accept,” or any numeric score above threshold, we replace it with “strong reject.” This tests whether the meta-review function uncritically adopts a contrarian verdict.

3.3 Rebuttal Perturbation Aspects

Rebuttal Tone Perturbation. We prompt GPT-4o to rewrite a rebuttal with heightened dismissiveness, sarcasm, or hostility toward the reviewers. For example, “We appreciate the reviewer’s suggestions” becomes “This reviewer’s suggestion offers no value.” This tests if an aggressively negative author response shifts the meta-review decision.

Rebuttal Presentation Perturbation. As with the paper-level presentation perturbation, we add typos, subject-verb mismatches, or disorganized paragraph structures to degrade readability. This tests whether an incoherent, disorganized rebuttal affects an LLM-based meta-review.

Rebuttal Completeness Perturbation. We remove substantive responses to major reviewer criticisms or replace them with generic placeholders (e.g., “We will consider it”), leaving only superficial replies. This creates ambiguous, unspecific rebuttals that test whether the meta-review is more likely to penalize incomplete author responses.

4 Experimental Design

Data Collection. We exclusively analyze *accepted* papers from ICLR 2024, guided by theoretical principles from perturbation-based robustness evaluation [23, 26]. Perturbation analysis theory underscores the importance of using texts of inherently high quality and methodological rigor as baselines; perturbing substandard or flawed inputs would introduce excessive noise, obscuring the meaningful effects of introduced perturbations [21, 23]. By selecting papers already validated through rigorous peer review, we ensure our perturbation results reflect genuine vulnerabilities within LLM-driven

peer-review workflows, rather than artifacts stemming from poor original quality. Using a stratified sample that mirrors official acceptance proportions (poster, spotlight, oral), we obtain 508 papers (406 posters, 83 spotlights, 19 orals), each with multiple reviews and a rebuttal. To preserve structural fidelity (e.g., formulas, tables), we rely on the ICLR 2024 parsed dataset by Yu et al. [29], which encodes math formulas, tables to $\LaTeX$ and keeps section hierarchies for reliable LLM processing.

Review Setup. To align our evaluation closely with real-world peer review practices, we primarily follow the ICLR guidelines³ to structure both *LLM-as-Reviewer* and *LLM-as-Meta-Reviewer* outputs. For *LLM-as-Reviewer*, we adopt the ICLR review template, including strengths, weaknesses, numeric ratings for three key dimensions (Contribution, Presentation, and Soundness; each rated from 1–4), and an overall recommendation score (1–10). Similarly, for *LLM-as-Meta-Reviewer*, we adhere to the ICLR meta-review guidelines, requiring the meta-reviewer to summarize reviewer discussions, provide balanced and nuanced evaluations, and generate an informed final recommendation. Specifically, we explore three distinct prompting configurations for meta-review generation: (1) None-CoT, outputting only a single overall score (1–10) and a final decision (Reject, Accept as Poster, Accept as Spotlight, Accept as Oral), without any justification; (2) Dimension-CoT, providing separate dimension scores for Contribution, Presentation, and Soundness (each rated from 1–4) as intermediate reasoning steps, along with a justified overall score (1–10) and final decision; (3) Template-CoT, generating a structured meta-review based on the standard ICLR meta-review template, including explicit sections for strengths and weaknesses synthesis (“Metareview”), justifications for why the paper did not receive a higher or lower score, an overall score (1–10), and a final decision. This setup allows us to examine how varying levels of explanation and structured prompting influence LLM vulnerability to manipulative inputs.

³<https://www.iclr.cc/Conferences/2024/ReviewerGuide>

LLMs. We adopt *GPT-4o* [1] as our primary large language model for both generating text perturbations (paper, review, and rebuttal) and producing the subsequent reviews or meta-reviews. Additionally, we evaluate *gemini-2.0-flash-001* [22], *Deepseek-chat-v3-0324* [4], *LLama3.3-70b-Instruct* [7], and *Mistral-small-3.1-24b-instruct* [9] on the same tasks, allowing for a comparative analysis of their responses under identical perturbation scenarios. All models are prompted in zero-shot setting⁴.

Statistical Tests. Our experiment encompasses two core methods for assessing performance shifts: 1) Directional Test: The *Directional Test* determines whether meaningful perturbations produce the expected *direction* of change in model outputs (e.g., lowered scores for more flawed content). Concretely, we employ the Wilcoxon Signed-Rank Test [25] to detect if the model’s scores significantly increase or decrease after perturbation. 2) Invariance Test: The *Invariance Test* checks whether irrelevant changes fail to produce significant differences, implying the model should yield *invariant* outputs if the core content remains unaffected. We adopt the Two One-Sided Tests (TOST) [12] procedure for equivalence.

5 LLM as Reviewer

In this section, We present the results of applying the perturbation framework to the *LLM-as-Reviewer* role. We analyze how targeted modifications to the input paper affect the LLM’s generated reviews and scores. Our analysis is structured around two key tests: a directional test (assessing whether perturbations cause changes in the expected direction), an invariance test (examining whether perturbations lead to changes where they should not), and a summary of key takeaways.

Table 1 shows the average differences in scores between the perturbed and baseline (unperturbed) conditions. Table 2 presents the results of statistical tests (both Wilcoxon Signed-Rank Test for directional changes and Two One-Sided Tests (TOST) for invariance) to determine the significance of the observed changes.

Table 1: Average score differences (perturbed - baseline) for LLM-as-Reviewer, by perturbation aspect (rows) and affected score (columns). Dimension scores: 1-4; Overall Rating: 1-10. Shading indicates magnitude; negative values are underlined.

Aspect	Contr.	Sound.	Prese.	Overall
Contribution	<u>-0.03</u>	<u>-0.02</u>	<u>-0.01</u>	<u>-0.04</u>
Soundness	<u>-0.02</u>	-0.00	-0.00	<u>-0.07</u>
Presentation	0.00	<u>-0.01</u>	-0.00	<u>-0.04</u>

5.1 Directional Test

The directional test evaluates whether modifications aimed at undermining specific aspects of the paper (Contribution, Soundness, Presentation) lead to statistically significant reductions in the corresponding scores and overall rating.

1) Contribution Perturbation: Exaggerating claims of novelty results in small average decreases across all scores (see Table 1),

⁴Following prior work [10, 29], we use a zero-shot setting due to the extensive input context length, which makes few-shot prompting impractical.

Table 2: Statistical test results for LLM-as-Reviewer. $\leftrightarrow$: No Significant Difference (NSD) + Equivalence (Equiv) (Wilcoxon & TOST); $\downarrow$: Significant decline (Wilcoxon). Equivalence margins: ± 0.5 (dimension scores), ± 1.0 (overall score). Cells shaded in **violet represent tests where perturbations are *expected to produce significant changes to this dimension score* (directional expectation), cells shaded in **teal** represent tests where perturbations are *expected to have no significant effect on this dimension score* (invariance expectation).**

Aspect Perturbed	Dimension Score	Result
Contribution	contribution_score	$\leftrightarrow$ (NSD + Equiv)
	soundness_score	$\leftrightarrow$ (NSD + Equiv)
	presentation_score	$\leftrightarrow$ (NSD + Equiv)
	overall_score	$\leftrightarrow$ (NSD + Equiv)
Presentation	presentation_score	$\leftrightarrow$ (NSD + Equiv)
	contribution_score	$\leftrightarrow$ (NSD + Equiv)
	soundness_score	$\leftrightarrow$ (NSD + Equiv)
	overall_score	$\leftrightarrow$ (NSD + Equiv)
Soundness	soundness_score	$\leftrightarrow$ (NSD + Equiv)
	contribution_score	$\leftrightarrow$ (NSD + Equiv)
	presentation_score	$\downarrow$ (before > after)
	overall_score	$\leftrightarrow$ (NSD + Equiv)

ranging from -0.01 to -0.04. Although all scores show a decrease on average, Table 2 indicates that none of these reductions are statistically significant, as they fall within the equivalence margin.

2) Soundness Perturbation: Omitting key methodological details leads to the largest average decrease in the overall rating (-0.07, as shown in Table 1), along with a significant decrease in the presentation score (refer to Table 2). However, the soundness score itself does not experience a significant decrease.

3) Presentation Perturbation: Introducing typos and disorganization results in a small average decrease in the overall score (-0.04) and in the soundness score (-0.01), as indicated in Table 1. Nonetheless, Table 2 reveals that these changes are also not statistically significant and remain within the equivalence margin.

5.2 Invariance Test

The invariance test examines whether perturbations to one aspect of the paper (e.g., presentation) inappropriately affect scores for other aspects (e.g., contribution, soundness). We expect a robust system to exhibit invariance: changes to presentation quality, for instance, should not significantly impact judgments of contribution or soundness. We use the Two One-Sided Tests (TOST) procedure to assess equivalence when determining no significant difference with the Wilcoxon test. Table 2 shows the results.

1) Contribution Perturbation: As anticipated, and confirmed by TOST, perturbations to the contribution aspect do not significantly affect the scores of other dimensions (all $\leftrightarrow$).

2) Presentation Perturbation: Similarly, superficial changes to presentation do not significantly alter scores of contribution or soundness, demonstrating appropriate invariance (all $\leftrightarrow$).

3) Soundness Perturbation: Critically, while the Soundness score itself shows no significant difference (and TOST confirms equivalence, $\leftrightarrow$), the Presentation score shows a statistically significant decrease ($\downarrow$). This violates the expected invariance: a flaw in soundness is being misattributed to an issue in presentation.

5.3 Takeaways

Based on the results above, the *LLM-as-Reviewer* experiments reveal several key vulnerabilities:

1) *Limited Sensitivity to Substantive Flaws* While the LLM penalizes some paper-level manipulations, these penalties are often mild, especially for issues related to contribution and presentation. The negative directional changes, though aligned with our expectation, are not evidential enough to show statistical significance.

2) *Misinterpretation of Missing Information* The most significant finding is that the LLM sometimes penalizes abstract methodological details without explanation (which is caused by the paper soundness perturbation) as presentation problems, rather than recognizing them as core soundness issues, which is a clear violation of expected invariance and reveals a fundamental misunderstanding of the different quality dimensions.

3) *Partial Robustness to Superficial Errors* The LLM does demonstrate appropriate invariance to superficial presentation errors when assessing contribution and soundness. This indicates some degree of robustness, but it is overshadowed by the more serious vulnerabilities.

These findings highlight the need for more sophisticated mechanisms to ensure that LLM-based reviewers accurately assess the core scientific contributions and methodological rigor of research papers and that they correctly attribute flaws to the appropriate quality dimensions.

6 LLM as Meta Reviewer

Before presenting our empirical results, we first clarify our expectations regarding the ideal responses of meta-reviewers to perturbations. A meta-reviewer processes three core components—papers, reviews, and rebuttals—but its primary responsibility is to assess the intrinsic quality of submitted papers. Thus, we anticipate that negative perturbations applied directly to papers or rebuttals, as they reflect genuine degradations in submission quality and author responses, should lead to meaningful decreases in meta-review scores. In contrast, perturbations to reviews—designed to mimic artificially malicious or overly negative reviewers—should ideally have minimal or no influence on the meta-review outcomes, especially given our deliberate selection of high-quality, *accepted* papers. An ideal meta-reviewer is expected to robustly recognize and discount such malicious reviews, ensuring the final evaluation accurately reflects the intrinsic merits of the submission, rather than reviewer bias.

With these expectations clearly defined, we proceed to analyze how targeted perturbations influence the LLM-generated meta-reviews, overall scores, and final decisions. The analysis is structured around two key tests: a directional test (assessing changes in

the expected direction) and an invariance test (examining changes where they should not occur), followed by a summary of key take-aways.

Table 3 presents the average differences in *LLM-as-Meta-Reviewer* overall score between the perturbed and baseline (unperturbed) inputs, and Figure 3 demonstrates the changes in acceptance rates after perturbation. By the Wilcoxon Signed-Rank Test and Two One-Sided Tests (TOST), we assess the significance of these differences of overall score and acceptance rate across various perturbation aspects and different chain-of-thought prompting styles, with the results in Table 4.

Table 3: Average overall score differences (perturbed - baseline) for LLM-as-Meta-Reviewer. Rows: Perturbation mode (Paper, Review, Rebuttal) and aspect. Columns: CoT prompt (Dimension, None, Template). Score range: 1-10. Underlined: negative; shading: magnitude.

Mode	Aspect	Dimension	None	Template
Paper	Contribution	0.08	0.34	0.47
	Soundness	0.09	0.36	0.48
	Presentation	0.06	0.32	0.47
Review	Tone	0.08	0.17	0.18
	Correctness	0.06	0.27	0.42
	Conclusion	<u>-0.01</u>	<u>-1.65</u>	<u>-0.58</u>
Rebuttal	Tone	0.07	0.09	0.17
	Completeness	0.09	0.14	0.15
	Presentation	0.08	0.09	0.16

6.1 Directional Test

The directional test assesses whether the perturbations contribute to directional changes in the LLM’s meta-review outputs. Specifically, we use the Wilcoxon Signed-Rank Test to determine if there are statistically significant differences in scores and shifts in final decision categories. We expect decreases in scores and a shift towards rejection when it comes to the perturbations of paper and rebuttals, since both of them are highly connected to degrading the quality of the paper as well as authors’ responses.

Paper Perturbations. 1) Overall Score: Contrary to expectations, all three paper-level perturbations (paper contribution, soundness, and presentation) generally lead to increases in the overall score assigned by the meta-reviewer (Table 3, Mode=Paper), and these increases are identified as statistically significant differences, indicated by the green up arrows in Table 4 (Mode=Paper, Overall Score).

2) Final Decision: Figure 3 and Table 4 (Mode=Paper, Final Decision) show the impact and the test results of the paper acceptance rate, respectively. Similar to the overall score results, we observe that paper perturbations often increase the likelihood of acceptance with statistical significance.

Rebuttal Perturbations. 1) Overall Score: Making the rebuttal incomplete (perturbation of rebuttal completeness), using a hostile tone (perturbation of rebuttal tones), or degrading clarity (perturbation of review presentation) of authors’ rebuttals paradoxically

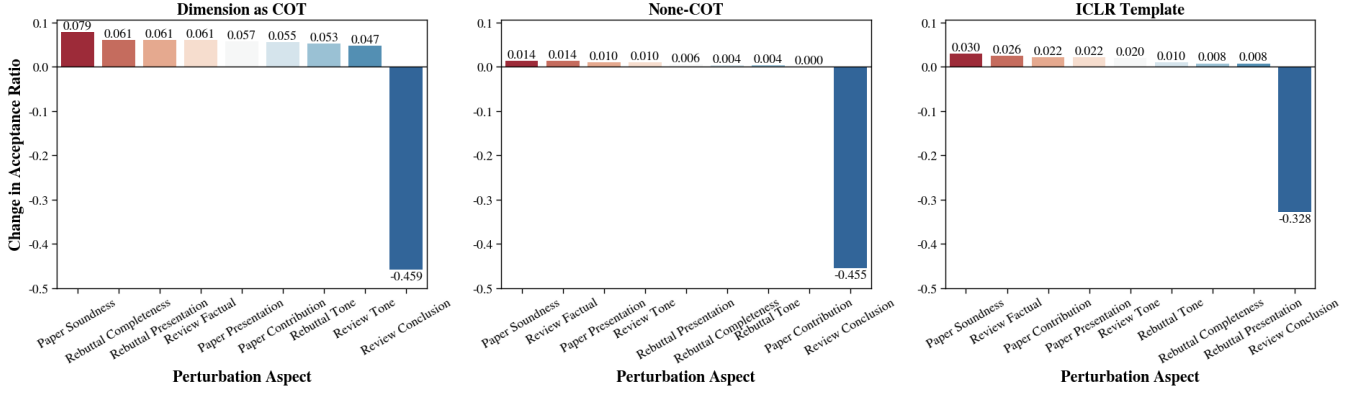


Figure 3: Percentage Difference in Acceptance Rate. Obtained and calculated from final decision, difference of acceptance rate in LLM-As-Meta-Reviewer outcomes ($Perturbed - Base$) across different aspects and chain-of thought variants (Dimension CoT, None CoT, ICLR Template CoT).

Table 4: Effects of Negative Perturbations on LLM-as-Meta-Reviewer Scores and Decisions. Rows specify perturbations applied (to Paper, Rebuttal, or Review), and columns indicate outcomes under different prompting conditions (Dimension, None, Template). Arrows represent statistically significant changes relative to the baseline (Wilcoxon Signed-Rank Test, $p < 0.05$): $\uparrow$ unexpected increase (indicating model vulnerability); $\downarrow$ appropriate decrease; $\leftrightarrow$ no significant change (robustness confirmed via TOST; margins: ± 0.5 Decision, ± 1.0 Score). All perturbations intentionally degrade content quality. Cells shaded in **violet** represent *Directional Tests*, where significant score or decision changes are expected because the perturbation directly affects core paper/rebuttal quality. Cells shaded in **teal** represent *Invariance Tests*, where no significant changes should occur, as the meta-reviewer is expected to robustly discount irrelevant or malicious review-level perturbations and maintain stable overall evaluations.

Mode	Aspect	Dimension		None		Template	
		Final	Overall	Final	Overall	Final	Overall
Paper	Contribution	$\uparrow$ Increase	$\uparrow$ Increase	$\leftrightarrow$ Invariance	$\uparrow$ Increase	$\uparrow$ Increase	$\uparrow$ Increase
	Presentation	$\uparrow$ Increase	$\uparrow$ Increase	$\leftrightarrow$ Invariance	$\uparrow$ Increase	$\uparrow$ Increase	$\uparrow$ Increase
	Soundness	$\uparrow$ Increase	$\uparrow$ Increase	$\uparrow$ Increase	$\uparrow$ Increase	$\uparrow$ Increase	$\uparrow$ Increase
Rebuttal	Completeness	$\uparrow$ Increase	$\uparrow$ Increase	$\leftrightarrow$ Invariance	$\uparrow$ Increase	$\leftrightarrow$ Invariance	$\uparrow$ Increase
	Presentation	$\uparrow$ Increase	$\uparrow$ Increase	$\leftrightarrow$ Invariance	$\uparrow$ Increase	$\uparrow$ Increase	$\uparrow$ Increase
	Tone	$\uparrow$ Increase	$\uparrow$ Increase	$\leftrightarrow$ Invariance	$\uparrow$ Increase	$\uparrow$ Increase	$\uparrow$ Increase
Review	Conclusion	$\downarrow$ Decrease	$\downarrow$ Decrease	$\downarrow$ Decrease	$\downarrow$ Decrease	$\downarrow$ Decrease	$\downarrow$ Decrease
	Factual	$\uparrow$ Increase	$\uparrow$ Increase	$\leftrightarrow$ Invariance	$\uparrow$ Increase	$\uparrow$ Increase	$\uparrow$ Increase
	Tone	$\uparrow$ Increase	$\uparrow$ Increase	$\leftrightarrow$ Invariance	$\uparrow$ Increase	$\uparrow$ Increase	$\uparrow$ Increase

increases the overall score assigned by the meta-reviewer in most cases with the significance confirmed in Table 4 (Mode=Rebuttal, Overall Score).

2) Final Decision: Figure 3 shows a similar increasing trend pattern of acceptance rate for rebuttal perturbation. Incomplete, hostile, or unclarified rebuttals may increase the probability of acceptance. This trend is also supported by the directional test result in Table 4 (Mode=Rebuttal, Final Decision), except for the None-CoT setting.

6.2 Invariance Test

The invariance test examines whether review-level perturbations to one component (tone, factual correctness, conclusion) inappropriately influence the meta-review outputs, since an ideal meta-reviewer is expected to be capable of recognizing malicious reviews and minimizing their impacts. For example, changing the tone of a review should not significantly alter the meta-reviewer’s assessment of the paper’s overall score and final decision.

We focus on whether the overall scores from Table 3 (mode=review) and the acceptance rate from Figure 3 (review perturbations) remain invariant when review-level aspects are perturbed. Note that the “Overall Score” and “Final Decision” are both expected to potentially

change under paper-level and rebuttal-level perturbations (e.g., an offensive rebuttal tone, while not directly related to soundness, could legitimately lower the overall assessment), which have been checked in the directional test and would not be discussed here.

Review Perturbations. 1) Overall Score: Flipping the reviewer’s conclusion to “Overall Score=1, strong reject” (perturbation of review conclusion) significantly decreases the overall score (Table 4, red down arrows, Mode=Review, Perturb Aspect=Conclusion, Overall Score). However, introducing a hostile tone (perturbation of review tone) or factual inaccuracies (perturbation of review factual correctness) often increases the overall score, which could be counterintuitive.

2) Final Decision: Consistent with the overall score, conclusion perturbation leads to a significant increase in rejections (Figure 3 and Table 4, Mode=Review, Perturb Aspect=Conclusion, Final Decision). However, the other review perturbations (tone, factual correctness) often increase acceptance rates, mirroring the unexpected findings for overall scores (except for situations under the None CoT prompt setting; despite the decreased acceptance rate shown in Figure 3, they remain statistically equivalent as we expected previously).

6.3 Takeaways

In conjunction with the above results, the *LLM-as-Meta-Reviewer* experiments show critical vulnerabilities:

1) *A Defect in Identifying Over-claims* The perturbation with regard to paper contribution is intended to degrade the paper score by adding over-claims or exaggerations about its contribution, whereas LLM shows insufficient ability to identify these over-claims. Instead of degrading, it goes overestimated scores and final decisions because of its recognition of these claims about paper contribution.

2) *Misinterpretation of Presentation* The perturbation of presentation (either paper-level or rebuttal-level) is designed to degrade the clarity of related texts, with an expectation of degrading scores and acceptance rates. However, LLM shows misinterpretation of this aspect. In some cases, LLM did notice the typos inserted by perturbation manipulation, while it judged them as sufficiently corrected according to authors’ responses, hence outputs higher scores because of its approval of authors’ corrections.

3) *Over-reliance on Overall Ratings* The meta-reviewer is heavily influenced by the reviewers’ overall numerical ratings, particularly strong rejections, often neglecting the detailed content within the reviews, resulting in unexpected changes under review conclusion perturbation.

4) *Inadequate Consideration for Tone and Completeness* Impolite tones and incomplete information (in rebuttals) are often less considered by LLM, leading to a meta-review with unbalanced weight across each aspect. In other words, though the weaknesses of tones or completeness in rebuttals are detected, LLM shows less attention to them and converges or somehow increases its ratings.

5) *Sensitivity to Prompt Settings Regarding CoT* LLMs are often sensitive to prompt settings. In this case, this sensitivity is evident in the divergent test results between the None-CoT and other CoT settings, despite sharing identical inputs. This suggests that a suitable prompt setting may offer partial safeguards against such variability.

These findings, taken together, demonstrate a fundamental lack of critical evaluation capabilities in the LLM-as-Meta-Reviewer. A probable explanation for the unexpected increase of score and acceptance rate is the preference of self-generated texts, according to some related works [19]. All underscore the urgent need for substantial improvements before reliably integrating these models into practical peer-review workflows.

7 Additional Results

7.1 Additional LLMs

We extend our evaluation to four additional models—*Gemini-2.0-flash-001*, *Deepseek-chat-v3-0324*, *LLaMA3.3-70B-Instruct*, and *Mistral-small-3.1-24B-Instruct*—to test the generality of our findings.

LLM as Reviewer. Table 5 shows average score shifts under perturbations. The pattern is consistent with our main results: Deepseek-V3 and LLaMA3.3-70B exhibit meaningful negative shifts across Contribution, Soundness, and Presentation, suggesting moderate ability to recognize degraded quality. In contrast, Gemini-2.0-flash and Mistral-3.1-24B show weak or inconsistent sensitivity, sometimes even increasing scores after perturbation. Overall, LLM reviewers demonstrate only partial robustness, with notable variance across models.

LLM as Meta-Reviewer. Table 6 reports meta-review outcomes. Strikingly, all four models replicate the core vulnerabilities identified with GPT-4o: (i) paper- and rebuttal-level degradations frequently *raise* overall scores rather than penalize them, and (ii) review-level manipulations, especially flipping reviewer conclusions, strongly sway final decisions. These findings reinforce that meta-reviewers are disproportionately influenced by surface-level cues (numeric recommendations, over-claims) and fail to discount misleading or malicious inputs.

Taken together, results across five families of LLMs confirm the generality of our diagnosis: current LLM reviewers are only partially reliable, and meta-reviewers remain especially vulnerable to biased or adversarial perturbations, underscoring the need for targeted safeguards in AI-assisted peer review.

7.2 Prompts and Case Studies

We demonstrate all prompts of the setting for LLM as Reviewer, LLM as Meta-Reviewer, and those related to our perturbations. Meanwhile, we demonstrate the performance of the perturbation attack across all designed aspects. To help readers better understand the difference between LLM outcomes before and after manipulations, we select five pairs of model outputs to be compared in detail, including LLM as Reviewer and LLM as Meta-Reviewer. For any readers interested in this part, please see the details in our anonymous link:

https://drive.google.com/file/d/1flvAJQ05bXukgUY072zOzkCiM45_5iaM/view?usp=drive_link

8 Related Work

LLMs in Peer Review. Modern Large Language Models have enabled automated or semi-automated pipelines for academic reviewing [10, 15, 29], already adopted in up to 15.8% of AI-conference reviews [13, 14]. While AI-generated comments can partially align

Table 5: Average differences in LLM-as-Reviewer scores (Perturbed – Base) across models and aspects. Dimension scores (Contribution, Soundness, Presentation) range from 1–4; Overall ratings (Ov.) range from 1–10. Negative values are underlined.

Model	Aspect	Contri.	Sound.	Present.	Ov.
Gemini-2.0-flash	Contribution	<u>-0.014</u>	<u>-0.010</u>	0.015	0.054
	Soundness	<u>-0.002</u>	<u>-0.010</u>	<u>-0.006</u>	0.062
	Presentation	0.014	<u>-0.006</u>	0.017	0.043
Deepseek-V3	Contribution	<u>-0.269</u>	<u>-0.261</u>	<u>-0.242</u>	<u>-0.242</u>
	Soundness	<u>-0.238</u>	<u>-0.277</u>	<u>-0.253</u>	<u>-0.253</u>
	Presentation	<u>-0.257</u>	<u>-0.263</u>	<u>-0.201</u>	<u>-0.236</u>
Llama3.3-70B	Contribution	<u>-0.242</u>	<u>-0.246</u>	<u>-0.234</u>	<u>-0.226</u>
	Soundness	<u>-0.244</u>	<u>-0.248</u>	<u>-0.244</u>	<u>-0.226</u>
	Presentation	<u>-0.230</u>	<u>-0.226</u>	<u>-0.222</u>	<u>-0.226</u>
Mistral-3.1-24B	Contribution	0.089	0.081	0.046	0.068
	Soundness	<u>-0.027</u>	0.004	0.027	0.135
	Presentation	0.019	<u>-0.052</u>	0.012	<u>-0.033</u>

with human judgments [15], concerns persist regarding hallucinations [30], biases [6], and susceptibility to adversarial inputs [15, 18]. Studies have also raised the risks of flawed critique interpretation, anonymity breaches, and undue reviewer influence in fully automated setups [28, 29].

Perturbation Analysis in NLP. Synthetic perturbations—ranging from minor lexical edits to deeper semantic shifts—serve as stress tests for model robustness across many NLP tasks [8, 11, 20]. Even subtle modifications, such as inserting factual errors or flipping a stance, can destabilize an LLM’s output [16, 31], particularly when models struggle to distinguish misleading from valid context [5, 30]. Although some peer-review studies apply single-level perturbations [28], most ignore how perturbations to the *review* or *rebuttal* also skew recommendations. Our approach broadens these evaluations by systematically altering all three elements (paper, review, rebuttal) and assessing LLM responses through both *directional* and *invariance* tests.

9 Conclusion

We proposed an *aspect-guided, multi-level perturbation framework* to systematically diagnose vulnerabilities in LLM-driven peer review. By perturbing specific aspects of *papers* (contribution, soundness, presentation), *reviews* (tone, correctness, conclusion), and *rebuttals* (tone, completeness, presentation), we revealed consistent weaknesses: LLM reviewers often misclassify methodological flaws as presentation issues, while meta-reviewers are overly swayed by strong rejections, incomplete or hostile rebuttals, and misleading critiques. These vulnerabilities persist across diverse prompting strategies and leading LLMs (GPT-4o, Gemini 2.0, LLaMa 3, Deepseek, Mistral).

Our contributions are threefold: (1) a structured taxonomy and dataset of adversarial perturbations; (2) empirical identification of review dimensions most prone to error; and (3) a reusable diagnostic toolkit for fine-grained vulnerability analysis. Together, these

Table 6: Average score differences (Perturbed – Baseline) for LLM-as-Meta-Reviewer across models, perturbation modes, and aspects. Ov.Dim, Ov.None, and Ov.Temp indicate overall scores under Dimension-CoT, None-CoT, and Template-CoT prompts (range: 1–10). Negative values are underlined.

Model	Mode	Aspect	Ov.Dim	Ov.None	Ov.Temp
Gemini-2.0-flash	Paper	Contribution	0.395	0.532	0.203
		Soundness	0.460	0.518	0.308
		Presentation	0.439	0.500	0.230
	Review	Factual	0.439	0.490	0.166
		Tone	<u>-0.050</u>	0.033	<u>-0.151</u>
		Conclusion	0.025	<u>-0.451</u>	<u>-0.141</u>
	Rebuttal	Completeness	<u>-0.077</u>	0.023	<u>-0.178</u>
		Presentation	<u>-0.110</u>	0.043	<u>-0.149</u>
		Tone	<u>-0.081</u>	0.052	<u>-0.172</u>
Deepseek-V3	Paper	Contribution	0.294	0.104	0.157
		Soundness	0.300	0.128	0.199
		Presentation	0.221	0.133	0.162
	Review	Factual	0.286	0.132	0.124
		Tone	0.188	<u>-0.052</u>	0.043
		Conclusion	<u>-0.033</u>	<u>-0.472</u>	<u>-0.191</u>
	Rebuttal	Completeness	0.188	<u>-0.087</u>	0.070
		Presentation	0.234	<u>-0.066</u>	0.116
		Tone	0.159	<u>-0.064</u>	0.023
Llama3.3-70B	Paper	Contribution	1.168	0.207	0.617
		Soundness	1.168	0.207	0.617
		Presentation	1.168	0.207	0.617
	Review	Factual	1.137	0.199	0.584
		Tone	1.064	0.199	0.528
		Conclusion	0.706	0.182	0.333
	Rebuttal	Completeness	1.064	0.199	0.528
		Presentation	1.064	0.199	0.528
		Tone	1.064	0.199	0.528
Mistral-3.1-24B	Paper	Contribution	0.387	0.698	0.166
		Soundness	0.424	0.700	0.217
		Presentation	0.404	0.706	0.219
	Review	Factual	0.423	0.662	0.224
		Tone	0.221	0.567	0.064
		Conclusion	0.108	0.524	<u>-0.112</u>
	Rebuttal	Completeness	0.193	0.540	0.087
		Presentation	0.238	0.495	0.143
		Tone	0.248	0.520	0.074

results provide actionable insights for designing robust, aspect-specific policies in human–AI peer review, clarifying where AI assistance is valuable and where human oversight is indispensable.

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In this work, we employed generative AI tools solely to polish the manuscript’s language. All content was carefully reviewed and validated manually to maintain accuracy and reliability.

References

- [1] Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altmenschmidt, Sam Altman, Shyamal Anadkat, et al. 2023. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774* (2023).
- [2] Balazs Aczel, Barnabas Szasz, and Alex O Holcombe. 2021. A billion-dollar donation: estimating the cost of researchers' time spent on peer review. *Research Integrity and Peer Review* 6 (2021), 1–8.
- [3] Lutz Bornmann and Rüdiger Mutz. 2015. Growth rates of modern science: A bibliometric analysis based on the number of publications and cited references. *Journal of the association for information science and technology* 66, 11 (2015), 2215–2222.
- [4] DeepSeek-AI, Aixin Liu, Bei Feng, Bing Xue, Bingxuan Wang, and et al. 2025. DeepSeek-V3 Technical Report. *arXiv:2412.19437 [cs.CL]* <https://arxiv.org/abs/2412.19437>
- [5] Ameet Deshpande, Vishvak Murahari, Tanmay Rajpurohit, Ashwin Kalyan, and Karthik Narasimhan. 2023. Toxicity in chatgpt: Analyzing persona-assigned language models. In *Findings of the Association for Computational Linguistics: EMNLP 2023*. 1236–1270.
- [6] Isabel O Gallegos, Ryan A Rossi, Joe Barrow, Md Mehrab Tanjim, Sunghul Kim, Franck Dernoncourt, Tong Yu, Ruiyi Zhang, and Nesreen K Ahmed. 2024. Bias and fairness in large language models: A survey. *Computational Linguistics* (2024), 1–79.
- [7] Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, and et al. 2024. The Llama 3 Herd of Models. *arXiv:2407.21783 [cs.AI]* <https://arxiv.org/abs/2407.21783>
- [8] Tianxing He, Jingyu Zhang, Tianle Wang, Sachin Kumar, Kyunghyun Cho, James R. Glass, and Yulia Tsvetkov. 2023. On the blind spots of model-based evaluation metrics for text generation. In *ACL (1)*. Association for Computational Linguistics, 12067–12097.
- [9] Albert Q Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, et al. 2023. Mistral 7B. *arXiv preprint arXiv:2310.06825* (2023).
- [10] Yiqiao Jin, Qinlin Zhao, Yiyang Wang, Hao Chen, Kaijie Zhu, Yijia Xiao, and Jindong Wang. 2024. AgentReview: Exploring Peer Review Dynamics with LLM Agents. *arXiv preprint arXiv:2406.12708* (2024).
- [11] Marzena Karpinska, Nishant Raj, Katherine Thai, Yixiao Song, Ankita Gupta, and Mohit Iyyer. 2022. DEMETR: diagnosing evaluation metrics for translation. In *EMNLP*. Association for Computational Linguistics, 9540–9561.
- [12] Daniël Lakens. 2017. Equivalence Tests: A Practical Primer for t Tests, Correlations, and Meta-Analyses. *Social Psychological and Personality Science* 8 (05 2017), 194855061769717. doi:10.1177/1948550617697177
- [13] Giuseppe Russo Latona, Manoel Horta Ribeiro, Tim R Davidson, Veniamin Veselovsky, and Robert West. 2024. The AI Review Lottery: Widespread AI-Assisted Peer Reviews Boost Paper Scores and Acceptance Rates. *arXiv preprint arXiv:2405.02150* (2024).
- [14] Weixin Liang, Zachary Izzo, Yaohui Zhang, Haley Lepp, Hancheng Cao, Xuan-dong Zhao, Lingjiao Chen, Haotian Ye, Sheng Liu, Zhi Huang, et al. 2024. Monitoring AI-Modified Content at Scale: A Case Study on the Impact of ChatGPT on AI Conference Peer Reviews. In *Forty-first International Conference on Machine Learning*.
- [15] Weixin Liang, Yuhui Zhang, Hancheng Cao, Binglu Wang, Daisy Yi Ding, Xinyu Yang, Kailas Vodrahalli, Siyu He, Daniel Scott Smith, Yian Yin, et al. 2024. Can large language models provide useful feedback on research papers? A large-scale empirical analysis. *NEJM AI* 1, 8 (2024), A10a2400196.
- [16] Yongkang Liu, Shi Feng, Daling Wang, Yifei Zhang, and Hinrich Schütze. 2023. Evaluate what you can't evaluate: Unassessable generated responses quality. <https://arxiv.org/abs/2305.14658>. CoRR, abs/2305.14658.
- [17] Chris Lu, Cong Lu, Robert Tjarko Lange, Jakob Foerster, Jeff Clune, and David Ha. 2024. The AI Scientist: Towards Fully Automated Open-Ended Scientific Discovery. *arXiv:2408.06292 [cs.AI]* <https://arxiv.org/abs/2408.06292>
- [18] Chris Lu, Cong Lu, Robert Tjarko Lange, Jakob Foerster, Jeff Clune, and David Ha. 2024. The ai scientist: Towards fully automated open-ended scientific discovery. *arXiv preprint arXiv:2408.06292* (2024).
- [19] Arjun Panickssery, Samuel R. Bowman, and Shi Feng. 2024. LLM Evaluators Recognize and Favor Their Own Generations. *arXiv:2404.13076 [cs.CL]* <https://arxiv.org/abs/2404.13076>
- [20] Ananya B. Sai, Tanay Dixit, Dev Yashpal Sheth, Sreyas Mohan, and Mitesh M. Khapra. 2021. Perturbation checklists for evaluating NLG evaluation metrics. In *EMNLP (1)*. Association for Computational Linguistics, 7219–7234.
- [21] A. Strauss and J.M. Corbin. 1998. *Basics of Qualitative Research: Techniques and Procedures for Developing Grounded Theory*. SAGE Publications. <https://books.google.de/books?id=wTwYUnHYsmMC>
- [22] Gemini Team, Rohan Anil, Sebastian Borgeaud, Jean-Baptiste Alayrac, and et al. 2024. Gemini: A Family of Highly Capable Multimodal Models. *arXiv:2312.11805 [cs.CL]* <https://arxiv.org/abs/2312.11805>
- [23] Florian Tramer, Nicholas Carlini, Wieland Brendel, and Aleksander Madry. 2020. On Adaptive Attacks to Adversarial Example Defenses. *arXiv:2002.08347 [cs.LG]* <https://arxiv.org/abs/2002.08347>
- [24] Keith Tyser, Ben Segev, Gaston Longhitano, Xin-Yu Zhang, Zachary Meeks, Jason Lee, Uday Garg, Nicholas Belsten, Avi Shporer, Madeleine Udell, et al. 2024. Ai-driven review systems: Evaluating llms in scalable and bias-aware academic reviews. *arXiv preprint arXiv:2408.10365* (2024).
- [25] Frank Wilcoxon. 1945. Individual Comparisons by Ranking Methods. *Biometrics Bulletin* 1, 6 (1945), 80–83. <http://www.jstor.org/stable/3001968>
- [26] Tongshuang Wu, Marco Tulio Ribeiro, Jeffrey Heer, and Daniel S. Weld. 2021. Polyjuice: Generating Counterfactuals for Explaining, Evaluating, and Improving Models. *arXiv:2101.00288 [cs.CL]* <https://arxiv.org/abs/2101.00288>
- [27] Yueqi Xie, Jingwei Yi, Jiawei Shao, Justin Curl, Lingjuan Lyu, Qifeng Chen, Xing Xie, and Fangzhao Wu. 2023. Defending chatgpt against jailbreak attack via self-reminders. *Nature Machine Intelligence* 5, 12 (2023), 1486–1496.
- [28] Rui Ye, Xianghe Pang, Jingyi Chai, Jiao Chen, Zhenfei Yin, Zhen Xiang, Xiaowen Dong, Jing Shao, and Siheng Chen. 2024. Are We There Yet? Revealing the Risks of Utilizing Large Language Models in Scholarly Peer Review. *arXiv preprint arXiv:2412.01708* (2024).
- [29] Jianxiang Yu, Zichen Ding, Jiaqi Tan, Kangyang Luo, Zhenmin Weng, Chenghua Gong, Long Zeng, Renjing Cui, Chengcheng Han, Qiushi Sun, et al. 2024. Automated Peer Reviewing in Paper SEA: Standardization, Evaluation, and Analysis. In *Findings of the Association for Computational Linguistics: EMNLP 2024*. 10164–10184.
- [30] Yi Zeng, Hongpeng Lin, Jingwen Zhang, Diyi Yang, Ruoxi Jia, and Weiyan Shi. 2024. How johnny can persuade llms to jailbreak them: Rethinking persuasion to challenge ai safety by humanizing llms. *arXiv preprint arXiv:2401.06373* (2024).
- [31] Chen Zhang, Luis Fernando D'Haro, Yiming Chen, Malu Zhang, and Haizhou Li. 2023. A comprehensive analysis of the effectiveness of large language models as automatic dialogue evaluators. <https://arxiv.org/abs/2312-15407>. CoRR, abs/2312-15407.